DO NOW — RECALL

Synoptic programming challenge

Name: _____ Class: _____ Date: _____

1. Give the Big O of a full scan of n values.

2. Which item does a queue remove first?

3. What does return count do?

4. Starting total=0, sum even values from [3,4,6].

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

O(n).

2. Model answer

The oldest enqueued item.

3. Model answer

Returns count to the caller.

4. Model answer

10 (4+6).

Vocabulary — know these terms

decomposition — Break a problem into smaller tasks.

threshold — Value used as a cutoff.

counter — Variable that records how often an event occurs.

boundary test — Input at or next to a permitted limit.

erroneous input — An input violating the program's constraints.

NEW LEARNING

Read, annotate and explain

Inputs and output

Inputs are an array and a threshold; the result is the number of qualifying items.

Check each element

The counter increases only for values greater than or equal to the threshold.

Test a boundary

A value equal to the threshold must count. For 5,8,3,8 at threshold 8, result is 2.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

Synoptic programming challenge

5 < 8	Skip	0

Synoptic programming challenge: 5 < 8, Skip, 0

New Learning

Synoptic programming challenge

5 < 8	Skip	0
8 ≥ 8	Count	1

Synoptic programming challenge: 5 < 8, Skip, 0; 8 ≥ 8, Count, 1

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

The second 6 and 8 count; final result is 3.

IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Count values at least 6 in [6,2,6,8].

The first 6 counts and 2 does not.

Count values at least 6 in [6,2,6,8].

6 ≥ 6	Count	1
2 < 6	Skip	1
6 ≥ 6	Count	2
8 ≥ 6	Count	3

Count values at least 6 in [6,2,6,8].: 6 ≥ 6, Count, 1; 2 < 6, Skip, 1; 6 ≥ 6, Count, 2; 8 ≥ 6, Count, 3

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

Synoptic programming challenge

5 < 8	Skip	0
8 ≥ 8	Count	1
3 < 8	Skip	1

Synoptic programming challenge: 5 < 8, Skip, 0; 8 ≥ 8, Count, 1; 3 < 8, Skip, 1

New Learning

Synoptic programming challenge

5 < 8	Skip	0
8 ≥ 8	Count	1
3 < 8	Skip	1
8 ≥ 8	Count	2

Synoptic programming challenge: 5 < 8, Skip, 0; 8 ≥ 8, Count, 1; 3 < 8, Skip, 1; 8 ≥ 8, Count, 2

I DO — WORKED EXAMPLE

Follow the worked method

Count values at least 7 in [4,7,9].

Predict

Skip 4 because it is below 7.

Explain

Count 7 and 9: the result is 2.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Count values at least 6 in [6,2,6,8].

The first 6 counts and 2 does not.

Count values at least 6 in [6,2,6,8].

6 ≥ 6	Count	1
2 < 6	Skip	1

Count values at least 6 in [6,2,6,8].: 6 ≥ 6, Count, 1; 2 < 6, Skip, 1

Count values at least 6 in [6,2,6,8].

The first 6 counts and 2 does not.

Count values at least 6 in [6,2,6,8].

6 ≥ 6	Count	1
2 < 6	Skip	1
6 ≥ 6	Count	2

Count values at least 6 in [6,2,6,8].: 6 ≥ 6, Count, 1; 2 < 6, Skip, 1; 6 ≥ 6, Count, 2

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The test score>=t excludes values equal to t.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Count values at least 7 in [4,7,9].

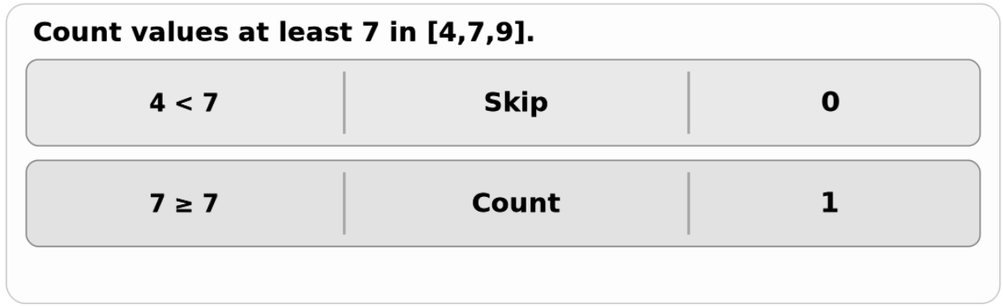
Skip 4 because it is below 7.



Count values at least 7 in [4,7,9].: 4 < 7, Skip, 0

Count values at least 7 in [4,7,9].

Skip 4 because it is below 7.



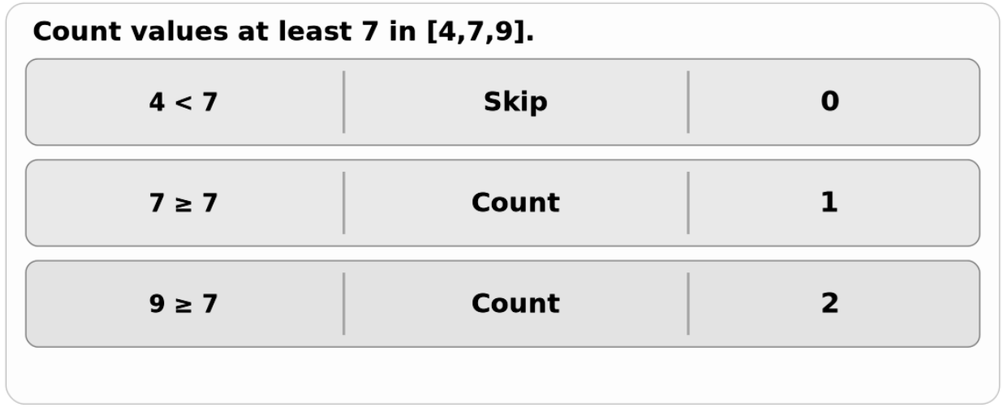
Count values at least 7 in [4,7,9].: 4 < 7, Skip, 0; 7 ≥ 7, Count, 1

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Count values at least 7 in [4,7,9].

Skip 4 because it is below 7.



Count values at least 7 in [4,7,9].: 4 < 7, Skip, 0; 7 ≥ 7, Count, 1; 9 ≥ 7, Count, 2

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Give normal, boundary and erroneous test cases if the threshold must be non-negative.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Find CountAtLeast([2,5,7],5).

Question 2

Change the method so it counts values strictly greater than the threshold.

WE DO — GUIDED PRACTICE

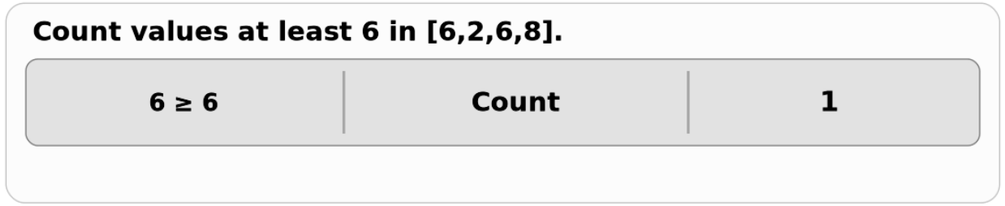
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Count values at least 6 in [6,2,6,8].

The first 6 counts and 2 does not.



Count values at least 6 in [6,2,6,8].: 6 ≥ 6, Count, 1

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict The first 6 counts and 2 does not.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.